

The empirical recurrence for OEIS A264566, recovered from its tail

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Abstract

OEIS A264566 records “Empirical recurrence of order 84” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 4096$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 85$ — every index at which it can be written down. Its last coefficient is $c_{84} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A264566 is “Number of $(n+1)X(5+1)$ arrays of permutations of $0..n*6+5$ with each element having directed index change 1,0 1,1 0,-1 or -1,1.” It has offset 1 and begins

2, 24, 143, 1130, 7808, 57265, 403872, 2873739, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Nov 17 2015 19:10:14, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 4096$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 4096$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 8192$ exact terms from $n = 1$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	6	c_{22}	−5886271153	c_{43}	1253659618755	c_{64}	−1110749449
c_2	5	c_{23}	9845294247	c_{44}	−1311293118109	c_{65}	929967772
c_3	71	c_{24}	−14584913303	c_{45}	897956270710	c_{66}	−267882048
c_4	−351	c_{25}	27471359051	c_{46}	−993864491766	c_{67}	123515968
c_5	458	c_{26}	−43083009448	c_{47}	977813894118	c_{68}	−109371233
c_6	−5087	c_{27}	56244822373	c_{48}	−606371310566	c_{69}	27783518
c_7	10696	c_{28}	−97768224642	c_{49}	617758772002	c_{70}	−9317178
c_8	−41904	c_{29}	144656793462	c_{50}	−574973254561	c_{71}	9164119
c_9	162108	c_{30}	−165942130976	c_{51}	321127666182	c_{72}	−2128698
c_{10}	−339356	c_{31}	269259144935	c_{52}	−300232469889	c_{73}	452561
c_{11}	1283957	c_{32}	−373150072288	c_{53}	266120263968	c_{74}	−527536
c_{12}	−3128284	c_{33}	380145828517	c_{54}	−132775501259	c_{75}	117682
c_{13}	7518025	c_{34}	−575774659708	c_{55}	113193371343	c_{76}	−13070
c_{14}	−21268844	c_{35}	742792850946	c_{56}	−96380836551	c_{77}	19681
c_{15}	42041515	c_{36}	−680163657637	c_{57}	42485728888	c_{78}	−4429
c_{16}	−102999952	c_{37}	956948600892	c_{58}	−32638880908	c_{79}	193
c_{17}	226661733	c_{38}	−1147791241335	c_{59}	27035662016	c_{80}	−425
c_{18}	−404014745	c_{39}	952091099747	c_{60}	−10398852572	c_{81}	100
c_{19}	930052499	c_{40}	−1239011957406	c_{61}	7050510609	c_{82}	−1
c_{20}	−1722656968	c_{41}	1384884693067	c_{62}	−5791563660	c_{83}	4
c_{21}	2830965615	c_{42}	−1043967191316	c_{63}	1926278304	c_{84}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 85$, and it is the recurrence OEIS A264566 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1 x^{84-1} - \dots - c_{84}$. Its constant term is $-c_{84} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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